

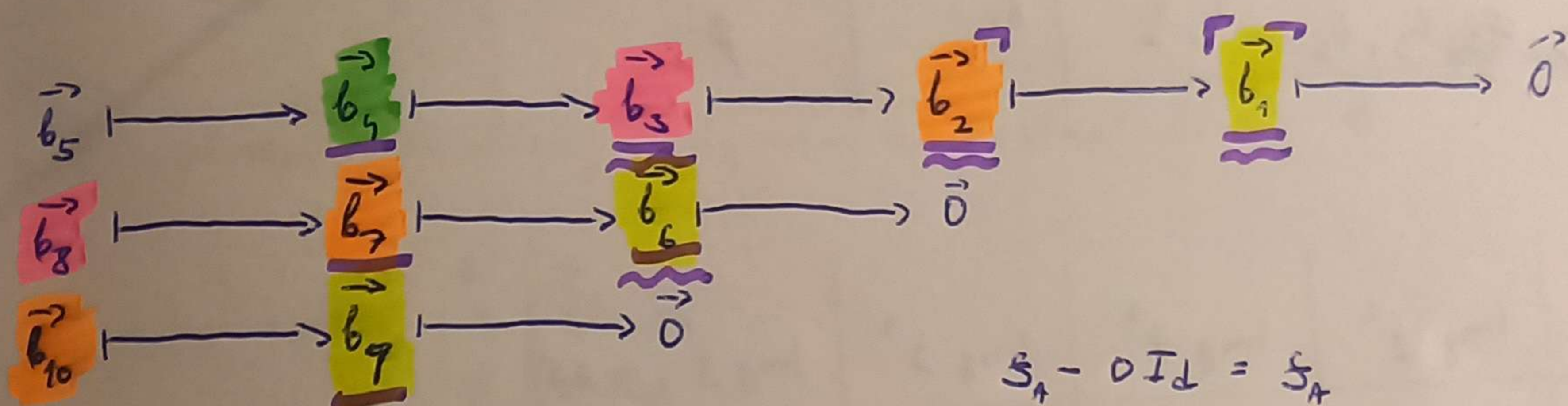
7.1.

$$[S]_B = \begin{pmatrix} J_{0,5} & 0 & 0 \\ 0 & J_{0,3} & 0 \\ 0 & 0 & J_{0,2} \end{pmatrix}$$

$J_{0,5} \rightarrow$ retiazok dĺžky 5 pre $\lambda = 0$

$J_{0,3} \rightarrow$ retiazok dĺžky 3 pre $\lambda = 0$

$J_{0,2} \rightarrow$ retiazok dĺžky 2 pre $\lambda = 0$



~~proje~~

$$\text{Ker } S = \text{Span} \{ \vec{b}_1, \vec{b}_6, \vec{b}_9 \}$$

$$\text{Ker } S^2 = \text{Span} \{ \vec{b}_1, \vec{b}_2, \vec{b}_6, \vec{b}_7, \vec{b}_9, \vec{b}_{10} \}$$

$$\text{Ker } S^3 = \text{Span} \{ \vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_6, \vec{b}_7, \vec{b}_8, \vec{b}_9, \vec{b}_{10} \}$$

$$\text{Ker } S^4 = \text{Span} \{ \vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_4, \vec{b}_6, \vec{b}_7, \vec{b}_8, \vec{b}_9, \vec{b}_{10} \}$$

$$\text{Ker } S^n; n \geq 5 = \text{Span} \{ B \}$$

$$\text{Im } S = \text{Span} \{ \vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_4, \vec{b}_6, \vec{b}_7, \vec{b}_9 \}$$

$$\text{Im } S^2 = \text{Span} \{ \vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_6 \}$$

$$\text{Im } S^3 = \text{Span} \{ \vec{b}_1, \vec{b}_2 \}$$

$$\text{Im } S^4 = \text{Span} \{ \vec{b}_1 \}$$

$$\text{Im } S^m; m \geq 5 = \emptyset$$

baza $\cap$	$\text{Im } f$	$\text{Im } f^2$	$\text{Im } f^3$	$\text{Im } f^4$	$\text{Im } f^m, m \geq 5$
$\text{Ker } f$	b_1, b_6, b_9	b_1, b_6	b_1	b_1	$\emptyset$
$\text{Ker } f^2$	b_1, b_6, b_2, b_7, b_9	b_1, b_2, b_6	b_1, b_2	b_1	$\emptyset$
$\text{Ker } f^3$	b_1, b_2, b_3 b_6, b_7, b_9	b_1, b_2 b_3, b_6	b_1, b_2	b_1	$\emptyset$
$\text{Ker } f^4$	b_1, b_2, b_3, b_4 b_6, b_7, b_9	b_1, b_2 b_3, b_6	b_1, b_2	b_1	$\emptyset$
$\text{Ker } f^n$ $n \geq 5$	$b_1, b_2, b_3, b_4, \cancel{b_5}$ $b_6, b_7, b_9, \cancel{b_{10}}$	b_1, b_2 b_3, b_6	b_1, b_2	b_1	$\emptyset$

$\dim(n)$	$\text{Im } f$	$\text{Im } f^2$	$\text{Im } f^3$	$\text{Im } f^4$	$\text{Im } f^m, m \geq 5$
$\text{Ker } f$	3	2	1	1	0
$\text{Ker } f^2$	5	3	2	1	0
$\text{Ker } f^3$	6	4	2	1	0
$\text{Ker } f^4$	7	4	2	1	0
$\text{Ker } f^n$ $n \geq 5$	10 7	4	2	1	0